

Polynomial-time optimal or approximate strategies for Pandora problem with budget-additive costs

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-27

Verifier verdict: APPROVED — progress score 3/3 (full solution)
Independent A.8 verifier assessment; pipeline details in the attribution footnote.

Abstract

We prove that it is NP-complete to decide whether a stochastic search instance with independent Bernoulli values and an explicitly represented budget-additive cost function has strictly positive optimal expected utility. The reduction is from Exact Cover by 3-Sets. Prime factorization encodes exact covers in the joint failure probabilities, while rational approximations to logarithms encode the corresponding inspection costs. A deterministic fallback box converts the resulting arithmetic separation into a utility gap.

The construction uses only rational probabilities, rewards, weights, and budgets of polynomial encoding length. After a polynomially encoded scaling, it yields an NP-hard promise gap between optimal value zero and optimal value at least one. Consequently, unless $P = NP$, there is no polynomial-time exact value algorithm, no value-reporting additive approximation with error strictly below $1/2$, and no finite-factor multiplicative approximation in the usual evaluable-output model.

1 The open problem

This paper addresses an open problem posed by Ben Berger, Tomer Ezra, Michal Feldman, Federico Fusco in *Pandoras Problem with Combinatorial Cost* [1] (Mathematics of Operations Research, 2024), Section 6 (Conclusion and Future Directions), page 25.. The website catalogue entry is problem 141 (W4405512051_p0) of the [Open Problems in Operations Research](#) collection.

Design an algorithm that, given oracle access to a budget-additive cost function $c(S) = \min\{B, \sum_{i \in S} w_i\}$ and explicit descriptions of independent value distributions $(D_i)_{i \in [n]}$, computes either:

- 1) *an optimal Pandora strategy $\pi^* \in \arg \max_{\pi} u(\pi)$, or*
- 2) *a strategy π with provable approximation guarantee for $\max_{\pi} u(\pi)$,*

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260827_142311_14a07996, sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

in time polynomial in n (and the input encoding / desired accuracy), using only polynomially many cost queries.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Model and statement of the result

There is a finite collection of boxes. Inspecting box i reveals an independent random value V_i . A strategy may choose boxes adaptively and may stop at any time. If S denotes the random set of inspected boxes, the strategy's expected utility is

$$\mathbb{E} \left[\max_{i \in S} V_i - c(S) \right],$$

where the maximum over the empty set is defined to be zero.

We consider explicitly represented budget-additive cost functions:

$$c(S) = \min \left\{ B, \sum_{i \in S} w_i \right\}, \quad B > 0, \quad w_i > 0.$$

Such a function is normalized, monotone, and submodular. Indeed, if $w(S) = \sum_{i \in S} w_i$, then the marginal cost of adding a box j to S is

$$c(S \cup \{j\}) - c(S) = \min\{w_j, (B - w(S))_+\},$$

which can only decrease as S grows.

A Bernoulli box takes value $v_i > 0$ with probability p_i and value zero otherwise. We also allow the degenerate case $p_i = 1$, so a deterministic positive-valued box is a Bernoulli box with support of size one.

An *impulsive strategy* fixes an order of some of the boxes and inspects them in that order until the first positive realization, at which point it stops. The empty impulsive strategy is allowed.

We use the following structural theorem from the source paper.

Theorem 1 (Impulsive optimality). *Every instance with independent Bernoulli box values and a submodular cost function has an optimal impulsive strategy.*

Our main result is the following.

Theorem 2. *It is NP-complete to decide whether*

$$\text{OPT} = \sup_{\pi} \mathbb{E} \left[\max_{i \in S_{\pi}} V_i - c(S_{\pi}) \right]$$

is strictly positive, even when all of the following hold:

1. *every box has support of size at most two;*

2. *the box values are mutually independent;*
3. *the cost function is explicitly represented and budget-additive; and*
4. *all probabilities, rewards, weights, and the budget are positive rationals of polynomial encoding length.*

The result is a computational lower bound for an explicit input representation; it is distinct from an information-theoretic value-oracle lower bound.

3 Reduction from Exact Cover by 3-Sets

Let

$$U = \{1, \dots, 3k\}$$

be the universe of an Exact Cover by 3-Sets (X3C) instance, and let

$$\mathcal{S} = \{S_1, \dots, S_m\}, \quad |S_i| = 3,$$

be its collection of triples. The question is whether there exists $A \subseteq \{1, \dots, m\}$ such that every element of U belongs to exactly one S_i with $i \in A$; such a set A is called an exact cover.

We may assume $k \geq 1$ and $m \geq k$. If $m < k$, the instance is trivially negative and can be mapped to any fixed negative search instance.

3.1 Prime encoding

Associate with each $e \in U$ a distinct prime r_e , taking $r_1, \dots, r_{3k}$ to be the first $3k$ primes. Define

$$P = \prod_{e \in U} r_e, \quad z_i = \prod_{e \in S_i} r_e.$$

For $A \subseteq \{1, \dots, m\}$, let

$$m_e(A) = |\{i \in A : e \in S_i\}|$$

and define

$$Z_A = \prod_{i \in A} z_i = \prod_{e \in U} r_e^{m_e(A)}.$$

Unique factorization gives the key equivalence

$$Z_A = P \iff m_e(A) = 1 \text{ for every } e \in U. \quad (1)$$

Thus $Z_A = P$ if and only if A is an exact cover.

Let

$$Z_{\max} = \prod_{i=1}^m z_i, \quad R = \max\{P, Z_{\max}\}.$$

Then, for every A ,

$$1 \leq Z_A \leq R, \quad \frac{1}{R} \leq \frac{Z_A}{P} \leq R. \quad (2)$$

Set

$$\Delta = \frac{1}{2R^5}, \quad \eta = \frac{\Delta}{4} = \frac{1}{8R^5}, \quad \delta = \frac{\Delta}{24m}.$$

For each $e \in U$, compute a positive rational number λ_e such that

$$|\lambda_e - \ln r_e| \leq \delta. \quad (3)$$

Define

$$L = \sum_{e \in U} \lambda_e, \quad K = L - 1, \quad C = \eta - \frac{1}{P}.$$

Since $R \geq P \geq 30$,

$$-\frac{1}{P} < C < 0.$$

The quantity C is an auxiliary negative rational; it is not an inspection weight, probability, or reward.

3.2 Boxes and costs

For each triple S_i , introduce an optional box i with cost weight

$$w_i = \sum_{e \in S_i} \lambda_e.$$

Let

$$q_i = \frac{1}{z_i}, \quad p_i = 1 - q_i,$$

and let the value of box i be

$$V_i = \begin{cases} v_i, & \text{with probability } p_i, \\ 0, & \text{with probability } q_i, \end{cases} \quad v_i = C + \frac{w_i}{p_i}.$$

Consequently,

$$p_i v_i - w_i = p_i C. \quad (4)$$

Write

$$W = \sum_{i=1}^m w_i,$$

and set the budget cap to

$$B = W + L.$$

Finally, introduce a fallback box f with cost weight

$$w_f = B$$

and deterministic value

$$H = B + C - K = W + 1 + C.$$

The complete cost function is therefore

$$c(T) = \min \left\{ B, \sum_{i \in T \cap \{1, \dots, m\}} w_i + B \mathbf{1}_{\{f \in T\}} \right\}. \quad (5)$$

Lemma 3. *The construction produces a valid budget-additive Bernoulli instance whose probabilities, rewards, weights, and budget are positive rationals of polynomial encoding length.*

Proof. Every S_i contains three distinct elements, so

$$z_i \geq 2 \cdot 3 \cdot 5 = 30 \quad \text{and} \quad p_i \geq \frac{29}{30}.$$

The precision δ is much smaller than one, so

$$w_i \geq \ln z_i - 3\delta \geq \ln 30 - 3\delta > 0.$$

Moreover,

$$C > -\frac{1}{P} \geq -\frac{1}{30}.$$

It follows that

$$v_i = C + \frac{w_i}{p_i} > 0.$$

Likewise,

$$H = W + 1 + C > 0,$$

and $B = W + L > W > 0$. Thus all actual instance parameters are positive. The fallback is a degenerate Bernoulli box with success probability one.

The first $3k$ primes have polynomial bit length and can be generated in polynomial time. Therefore P , the z_i , $Z_{\max}$, and R have polynomial bit length. Furthermore,

$$\log \frac{1}{\delta} = O(\log m + \log R).$$

Natural logarithms of integers can be approximated by dyadic rationals to this precision in time polynomial in the input length and the requested number of precision bits. Thus the λ_e , and therefore w_i , L , K , B , H , and v_i , all have polynomial encoding length. Finally, (5) is explicitly budget-additive and hence normalized, monotone, and submodular. $\square$

4 The arithmetic separation

Define

$$\phi(Z) = \frac{\ln Z - \ln P + 1}{Z},$$

and note that

$$\phi(P) = \frac{1}{P}.$$

Lemma 4 (Separation at P). *Let Z be an integer satisfying*

$$1 \leq Z \leq R \quad \text{and} \quad \frac{1}{R} \leq \frac{Z}{P} \leq R.$$

If $Z \neq P$, then

$$\frac{1}{P} - \phi(Z) \geq \Delta.$$

Proof. Set $x = Z/P$. A direct calculation gives

$$\frac{1}{P} - \frac{\ln Z - \ln P + 1}{Z} = \frac{x - 1 - \ln x}{Z}. \quad (6)$$

Let

$$h(x) = x - 1 - \ln x.$$

On the interval $[1/R, R]$,

$$h''(x) = \frac{1}{x^2} \geq \frac{1}{R^2}.$$

Since $h(1) = h'(1) = 0$, strong convexity gives

$$h(x) \geq \frac{(x - 1)^2}{2R^2}.$$

Because Z and P are distinct integers,

$$|x - 1| = \frac{|Z - P|}{P} \geq \frac{1}{P} \geq \frac{1}{R}.$$

Using also $Z \leq R$, equation (6) yields

$$\frac{1}{P} - \phi(Z) = \frac{h(x)}{Z} \geq \frac{1}{2R^2} \cdot \frac{1}{R^2} \cdot \frac{1}{R} = \frac{1}{2R^5} = \Delta. \quad \square$$

For $A \subseteq \{1, \dots, m\}$, let

$$s_A = \sum_{i \in A} w_i.$$

Lemma 5 (Rational approximation error). *For every $A \subseteq \{1, \dots, m\}$,*

$$\left| \frac{s_A - K}{Z_A} - \phi(Z_A) \right| \leq \frac{\Delta}{4}.$$

Proof. From the definitions,

$$s_A = \sum_{e \in U} m_e(A) \lambda_e$$

and

$$s_A - K = 1 + \sum_{e \in U} (m_e(A) - 1) \lambda_e.$$

On the other hand,

$$\ln Z_A - \ln P + 1 = 1 + \sum_{e \in U} (m_e(A) - 1) \ln r_e.$$

Consequently,

$$\frac{s_A - K}{Z_A} - \phi(Z_A) = \frac{\sum_{e \in U} (m_e(A) - 1) (\lambda_e - \ln r_e)}{Z_A}.$$

Since $Z_A \geq 1$, equation (3) gives

$$\left| \frac{s_A - K}{Z_A} - \phi(Z_A) \right| \leq \delta \sum_{e \in U} |m_e(A) - 1|.$$

Furthermore,

$$\sum_{e \in U} |m_e(A) - 1| \leq \sum_{e \in U} (m_e(A) + 1) = 3|A| + 3k \leq 6m,$$

where the last inequality uses $|A| \leq m$ and $k \leq m$. Therefore

$$\left| \frac{s_A - K}{Z_A} - \phi(Z_A) \right| \leq 6m\delta = \frac{\Delta}{4}. \quad \square$$

If A is an exact cover, then $m_e(A) = 1$ for every $e \in U$, and hence

$$s_A = L = K + 1. \quad (7)$$

5 Utilities of impulsive strategies

By Theorem 1, the constructed instance has an optimal impulsive strategy. Because the fallback succeeds with probability one, any boxes placed after it are unreachable. Thus every relevant impulsive strategy has one of two forms:

1. an ordered collection of optional boxes, with no fallback; or
2. an ordered collection of optional boxes followed by the fallback.

Let $A = (i_1, \dots, i_t)$ be an ordering of distinct optional boxes. For $1 \leq \ell \leq t$, let

$$Q_{\ell-1} = \prod_{r < \ell} q_{i_r}$$

be the probability that box i_ℓ is reached. Also write

$$Q_A = \prod_{i \in A} q_i = \frac{1}{Z_A}.$$

Since $B > W$, the budget cap never binds while only optional boxes have been inspected. Thus the marginal cost of optional box i is exactly w_i .

Lemma 6 (Optional strategies). *The utility of an impulsive strategy consisting only of the optional boxes in A is*

$$U_{\text{opt}}(A) = C(1 - Q_A).$$

It is zero for $A = \emptyset$ and strictly negative for every nonempty A .

Proof. Using (4),

$$\begin{aligned} U_{\text{opt}}(A) &= \sum_{\ell=1}^t Q_{\ell-1} (p_{i_\ell} v_{i_\ell} - w_{i_\ell}) \\ &= C \sum_{\ell=1}^t Q_{\ell-1} p_{i_\ell}. \end{aligned}$$

Since $p_i = 1 - q_i$, the last sum telescopes:

$$\sum_{\ell=1}^t Q_{\ell-1} p_{i_\ell} = 1 - \prod_{i \in A} q_i = 1 - Q_A.$$

Thus

$$U_{\text{opt}}(A) = C(1 - Q_A),$$

and the conclusion follows from $C < 0$. $\square$

Lemma 7 (Fallback strategies). *If the fallback is inspected after the optional boxes in A , the resulting utility is*

$$U_f(A) = C + \frac{s_A - K}{Z_A}.$$

In particular, the utility depends only on A , not on its order.

Proof. The fallback is reached precisely when every optional box in A fails, which has probability

$$Q_A = \frac{1}{Z_A}.$$

Before the fallback, the accumulated cost is $s_A < B$. Since the fallback has cost weight B , inspecting it raises the total cost to the cap B , so its marginal cost is exactly

$$B - s_A.$$

Combining the optional-box contribution from Lemma 6 with the fallback contribution gives

$$\begin{aligned} U_f(A) &= C(1 - Q_A) + Q_A(H - (B - s_A)) \\ &= C(1 - Q_A) + Q_A(H - B + s_A). \end{aligned}$$

Since $H = B + C - K$,

$$U_f(A) = C(1 - Q_A) + Q_A(C - K + s_A) = C + Q_A(s_A - K).$$

Substituting $Q_A = 1/Z_A$ proves the claim. $\square$

6 Correctness of the reduction

Lemma 8 (Exact-cover case). *If the $X3C$ instance has an exact cover A , then the constructed search instance satisfies*

$$\text{OPT} \geq \eta > 0.$$

Proof. For an exact cover, equations (1) and (7) give

$$Z_A = P, \quad s_A - K = 1.$$

Therefore Lemma 7 yields

$$U_f(A) = C + \frac{1}{P} = \eta. \quad \square$$

Lemma 9 (Non-cover case). *If A is not an exact cover, then*

$$U_f(A) \leq -\frac{\Delta}{2}.$$

Proof. By unique factorization, $Z_A \neq P$. Lemmas 4 and 5 imply

$$\frac{s_A - K}{Z_A} \leq \phi(Z_A) + \frac{\Delta}{4} \leq \frac{1}{P} - \Delta + \frac{\Delta}{4}.$$

Using $C = \eta - 1/P$ and $\eta = \Delta/4$, we obtain

$$\begin{aligned} U_f(A) &= C + \frac{s_A - K}{Z_A} \\ &\leq \eta - \frac{1}{P} + \frac{1}{P} - \Delta + \frac{\Delta}{4} \\ &= \frac{\Delta}{4} - \Delta + \frac{\Delta}{4} \\ &= -\frac{\Delta}{2}. \end{aligned}$$

□

We can now prove the main theorem.

Proof of Theorem 2. We first prove NP-hardness. If the X3C instance is positive, Lemma 8 gives a strategy of utility $\eta > 0$, so $\text{OPT} > 0$.

Suppose instead that the X3C instance is negative. Every impulsive strategy without the fallback has nonpositive utility by Lemma 6. Every impulsive strategy containing the fallback is, after removing unreachable boxes, a sequence of optional boxes followed by the fallback; its optional set is not an exact cover, so Lemma 9 shows that its utility is strictly negative. The empty strategy has utility zero.

Theorem 1 guarantees the existence of a globally optimal impulsive strategy. Hence, in the negative X3C case,

$$\text{OPT} = 0.$$

We have therefore established

$$\text{OPT} > 0 \iff \text{the X3C instance has an exact cover.}$$

The construction is polynomial-time by Lemma 3, which proves NP-hardness.

It remains to prove membership in NP. An impulsive order is a polynomial-size certificate. For an order $(i_1, \dots, i_t)$, let

$$T_\ell = \{i_1, \dots, i_\ell\} \quad (\text{with } T_0 = \emptyset), \quad Q_{\ell-1} = \prod_{r < \ell} (1 - p_{i_r}).$$

Its utility is exactly

$$\sum_{\ell=1}^t Q_{\ell-1} (p_{i_\ell} v_{i_\ell} - (c(T_\ell) - c(T_{\ell-1}))).$$

All terms are rational, and their numerators and denominators retain polynomial bit length. The utility can therefore be evaluated and compared with zero in polynomial time. By Theorem 1, an instance with positive optimal value has a positive-utility impulsive certificate. Thus the decision problem belongs to NP. □

7 Approximation consequences

The preceding construction guarantees a positive-case utility of only η , but its encoding length is polynomial, so a scaling turns it into a unit gap.

Corollary 10. *There is a polynomial-time reduction from X3C producing instances satisfying the promise*

$$\text{OPT} = 0 \quad \text{or} \quad \text{OPT} \geq 1.$$

Proof. Let

$$\alpha = \eta^{-1} = 8R^5.$$

Multiply every reward, every cost weight, and the budget cap by α , leaving all probabilities unchanged. The new cost function is

$$c'(S) = \min \left\{ \alpha B, \sum_{i \in S} \alpha w_i \right\} = \alpha c(S).$$

Likewise,

$$\max_{i \in S} \alpha V_i - \alpha c(S) = \alpha \left(\max_{i \in S} V_i - c(S) \right).$$

Thus every strategy's utility, and hence the optimal utility, is multiplied by α . A negative X3C instance still has optimal value zero, while a positive instance has optimal value at least $\alpha\eta = 1$.

Because $\alpha = 8R^5$ has $O(\log R)$ bits, the scaling preserves polynomial encoding length. $\square$

Corollary 11. *Unless $P = NP$, none of the following exists:*

1. *a polynomial-time algorithm that reports OPT exactly;*
2. *a polynomial-time value algorithm with additive error strictly smaller than 1/2;*
3. *a polynomial-time finite-factor multiplicative approximation in the standard value-reporting model; or*
4. *a polynomial-time algorithm returning an optimal, or finite-factor approximate, strategy whose expected utility is polynomial-time evaluable.*

The final statement applies, in particular, to algorithms returning impulsive or fixed-order strategies.

Proof. Apply the algorithms to the scaled instances from Corollary 10. An additive estimate with error less than 1/2 distinguishes the two cases by comparison with 1/2.

For any finite multiplicative factor ρ , a strategy satisfying

$$U(\pi) \geq \frac{\text{OPT}}{\rho}$$

has strictly positive utility in every positive instance, whereas in a zero-optimum instance no strategy has positive utility. Thus the sign of an exactly evaluable returned strategy distinguishes the two cases. The same zero-versus-positive distinction applies to the usual multiplicative value-reporting guarantee. Any of these algorithms would therefore decide X3C in polynomial time. $\square$

Remark 12. The qualification concerning evaluable strategy outputs is necessary: an arbitrary representation of a highly adaptive strategy need not come with a polynomial-time procedure for evaluating its expected utility. No such qualification is needed for exact or approximate value-reporting algorithms. Impulsive and fixed-order outputs are exactly evaluable using rational prefix-failure probabilities.

8 Conclusion

The construction shows that the difficulty of budget-additive stochastic search is computational rather than merely information-theoretic. Even though the cost function has a succinct explicit representation, determining whether any strategy improves strictly upon the empty strategy is NP-complete.

The reduction uses the structural existence of an optimal impulsive strategy only to classify an optimal policy. Prime factorization then encodes exact cover in the probability that all optional boxes fail, and the logarithmic weights make the fallback contribution attain its unique positive target precisely at an exact cover. This resolves the polynomial-time optimization question negatively under $P \neq NP$.

References

- [1] Ben Berger, Tomer Ezra, Michal Feldman, Federico Fusco. *Pandoras Problem with Combinatorial Cost*. Mathematics of Operations Research, 2024. <https://doi.org/10.1287/moor.2023.0248>